Score Agreement Between AI-Based and Rule-Based Resume–Job Matching: An Exploratory Evaluation on Synthetic Cases

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Abstract

Purpose. Resume–job matching tools increasingly combine a deterministic rule-based scorer with a large language model (LLM) scorer, each emitting a nominally commensurate 0–100 score. We ask how closely the two scorers agree on identical inputs and report an exploratory evaluation in one such application.

Methods. We froze a fully synthetic benchmark (12 resumes × 10 job descriptions, 120 pairs), a seeded run order, and a protocol fixing the agreement metrics before analysis. The AI arm used one served model (openai/gpt-oss-120b, temperature 0.1) with one fixed prompt; the rule arm is a deterministic scorer recomputed offline. Agreement statistics (Bland–Altman) were computed on the first 31 pairs of the frozen random order.

Results. The mean absolute score difference — the average absolute difference between the AI score and the rule-based score across the 31 observations — was 9.13 points; the median absolute difference was 8 points; the mean bias was +4.16 points; the standard deviation (SD) of differences was 10.49 points; and limits of agreement were −16.40 to +24.72 points. 38.71% of pairs lay within ±5 points and 67.74% within ±10 points. Pearson $r = 0.91$ and Spearman $\rho = 0.91$ indicate strong association, not agreement; with no ground-truth scores, accuracy is not assessed. Planned repeatability (0/12 runs) and controlled-edit (0/3, 0/16 runs) experiments were not executed.

Conclusion. On this panel the scorers ordered pairs similarly yet differed by up to 24 points on individual pairs. Evaluations of dual-scorer pipelines should report absolute score differences alongside correlation.

Keywords: resume–job matching; large language models; rule-based systems; score agreement; Bland–Altman analysis; synthetic benchmark

1. Introduction

1.1 Background and problem

Automated resume–job matching is a component of applicant-tracking and career-guidance software. A common architecture pairs a deterministic scorer — a keyword and feature engine whose score follows fixed rules from extracted skills, required-skill overlap, experience, and education — with a scorer based on a large language model (LLM), which returns a 0–100 fit score together with a natural-language explanation. Because both numbers are presented on the same scale and often displayed side by side to recruiters or candidates, an operator faces a natural question that is rarely measured: on the same inputs, how closely do the two scorers' numbers agree?

Two neighbouring questions belong to the same operational concern: whether the LLM scorer's number is stable when the same pair is scored again under fixed conditions, and whether it moves when a resume is edited in a controlled way. All three questions were pre-specified in the study protocol; this paper answers the first one only, because the other two were never executed (Section 1.2).

The literature reviewed in our targeted search covers adjacent questions: supervised prediction of person–job fit against labelled data [1–3], comparison of LLM-based screening with human raters [4], audits of demographic bias under controlled identity edits [5], multi-agent and zero-shot LLM screening systems [6, 7], robustness to strategic manipulation of resumes [8], controlled attribute manipulation across models [9], and the sensitivity of LLM outputs to prompt formatting [10]. The gap this paper addresses is narrower than the protocol's full design: among the studies reviewed in our targeted search we did not identify a study that compares score-level agreement between an LLM-based and a rule-based resume–job matcher on identical inputs within one deployed application. The protocol's further ambition — adding fixed-condition repeatability and controlled-edit sensitivity to the same comparison — is likewise unoccupied in the reviewed set, but that combination was never measured, so it is claimed here only as a pre-specified extension, not as a contribution. This paper reports the completed agreement comparison (RQ1); the other two components are unmeasured future work (Section 1.2).

1.2 Research questions and the scope of this paper

The study protocol specified three research questions for the CareerLens application before analysis:

- **RQ1 (score agreement):** How closely do the AI-based and rule-based CareerLens matchers' scores agree across identical resume–job pairs?
- **RQ2 (repeatability):** How consistent are AI-matcher scores across independent runs under documented, fixed conditions?
- **RQ3 (sensitivity):** How do the scores produced by each matcher change after controlled, role-relevant and irrelevant resume edits?

Two questions were explicitly excluded from the outset. No accuracy question is posed: no independently validated ground-truth scores exist for these pairs, and agreement with one scorer is not evidence of accuracy [11]. No hiring-bias question is posed: that would require identity manipulation and an ethics pathway; the bias audits [5, 9] are cited as context only.

Because only the RQ1 observations were collected, this manuscript reports an RQ1-only study. The original protocol, its three-question design, and its provisional three-question title are unchanged historical records. The reporting scope was narrowed after the study was initiated (but before any analysis was run), because the remaining observations were never collected: RQ2 has 0 of its 12 planned repeat runs, and RQ3 has 0 of its 3 planned baselines and 0 of its 16 planned controlled-edit runs. This narrowing is disclosed here and in Sections 3.3, 4.3, and 6. RQ2

and RQ3 are described only as unmeasured future work, never as results, and the absence of those data is not evidence about the questions they would answer.

The protocol also recorded directional expectations, frozen before analysis: for RQ1, a positive association between the two arms' scores (direction only); for RQ2, greater-than-zero within-pair variability for the LLM arm and exactly zero for the deterministic rule arm; for RQ3, strictly directional rule-score changes for role-relevant edits and exactly zero rule-score change for irrelevant edits, with the LLM arm expected to follow in direction. Only the RQ1 expectation bears on this paper, and the panel results in Section 4.2 are consistent with it; no magnitude was predicted and none is claimed beyond the panel.

1.3 Contributions

This paper makes one empirical contribution and two supporting contributions, stated at the scale the evidence supports:

1. **Empirical (central).** An exploratory measurement of score agreement between an LLM scorer and an explicit rule-based scorer on a shared 0–100 scale, on identical resume–job inputs within one deployed application, reported with absolute-difference statistics, indicative uncertainty, and a complete accounting of failed and superseded model calls.

2. **Artifacts.** A frozen, fully synthetic benchmark and a dated analysis protocol, together with a checksummed local reproducibility package, that leave the unexecuted repeatability and sensitivity studies precisely specified and reproducible by others.

3. **Reporting.** A transparent report that separates what was measured (RQ1) from what was pre-specified but not measured (RQ2: 0/12; RQ3: 0/3 and 0/16), so that readers cannot mistake planned work for evidence, or the lack of data for a negative finding.

This paper does not claim accuracy, fairness, deployment readiness, or generalisability, and it makes no claim about the relative quality of the two scorers; it measures correspondence, not correctness.

2. Related work

The review summarised here is targeted rather than systematic; every cited source was verified against a DOI registration record or an official publisher page. The qualified gap statement in Section 1.1 is the only absence claim made.

2.1 Person–job fit prediction

Zhu et al. [1] learn person–job fit end-to-end with item-level explanations on large application data; Qin et al. [2] model per-requirement ability explicitly against keyword and feature baselines of the same family as the rule scorer studied here; Wang et al. [3] add co-attention over candidate profile and recruitment history. These systems optimise prediction against labelled outcomes, not agreement between two deployed scorers.

2.2 LLM screening compared with human raters

Vaishampayan et al. [4] compared zero-shot GPT-4 with human ratings on 736 real resumes and reported minimal correlation between LLM and human scores, concluding that the two are not interchangeable; chain-of-thought prompting improved rating quality. Lo et al. [6] reported a retrieval-augmented multi-agent screener whose scores correlate with HR ratings (PC10 0.84 with DeepSeek-V3), and Skondras et al. [7] built a zero-shot matcher using structured prompting and semantic embeddings. These studies compare an LLM against humans on real resumes; none compares an LLM scorer with a rule-based scorer on a shared numeric scale, and none includes a repeated-run arm.

2.3 Controlled audits of bias and strategic manipulation

An et al. [5] randomised social identities across approximately 361,000 resumes and five LLM families; Iso et al. [9] manipulated names and institutions across many models with ROC-AUC and bias regression; Cohen et al. [8] studied strategic manipulation of resumes against an open LLM screening tool. Together they establish controlled manipulation as a publishable audit method, but their manipulated attributes are identities and strategy rather than role-relevant and role-irrelevant resume content, and they include no deterministic control arm.

2.4 Evaluation methodology

Raghavan et al. [11] document why evaluation claims in algorithmic hiring need validity evidence, anchoring the caution that a score is not a validated measure. Sclar et al. [10] showed that meaning-preserving prompt-format changes can swing LLM accuracy by up to 76 points and recommend reporting ranges over formats rather than single numbers; this study adopts the analogous discipline of reporting the distribution of pairwise score differences rather than a single summary, while declaring its single fixed prompt as a limitation. Dadaboyev et al. [12] systematically review AI in recruitment and motivate controlled evaluation over product demonstration.

2.5 Cross-system comparison and repeatability

Vásquez-Rodríguez et al. [13] compared human, rule-based, LLM, and supervised skill-extraction outputs — the closest cross-system design in our search — but on extraction text similarity rather than numeric match scores, and without repeated runs or controlled edits. Varshney and Ganuthula [14] (a preprint) tested LLM resume-screening repeatability across contextual variations with ANOVA and paired tests, finding significant between-model and model-versus-human differences on identical resumes; this is the closest design to RQ2, but it varies models and contexts rather than isolating fixed-condition within-pair variability against a deterministic control. Castleman et al. [15] audit LLM resume rankers for validity — including abstention on equally qualified pairs — against constructed ground-truth-superiority datasets; this is the closest work to the validity framing used here, but it is a single-system study with no rule-versus-LLM shared-scale comparison and no repeated-run arm.

In summary, among the studies reviewed in our targeted search, none occupies the joint cell of LLM-versus-rule comparison on a shared score, fixed-condition repeated runs, and controlled content edits within one application; the present paper occupies only the first of those three dimensions and reports it as the completed contribution, while the other two remain specified but unmeasured. Unlike the human-rating studies [4, 6, 7], both systems compared here are automated and see identical inputs; unlike the constructed-ground-truth audit [15], we measure correspondence between two scorers rather than validity against labels — the construct is agreement, not accuracy.

3. Methodology

3.1 Systems under test

The CareerLens dual-scorer pipeline scores a resume–job pair twice. The **rule arm** (`backend/ruleBasedFallback.js`, function `buildMatchAnalysis`) computes a 0–100 score deterministically from extracted skills, required-skill matching, keyword-band overlap, experience, and education components; identical inputs always yield an identical score. The **AI arm** (`analyzeMatch` → `aiProvider.js` → `llm.js`) sends the resume and job description to a served LLM under one fixed production prompt (resume and job text clipped to 14k and 8k characters; temperature 0.1; 2000-token output budget) and returns a 0–100 score with a breakdown and a summary.

One recorded deviation applies to this study: the AI arm was served by a single provider (Groq, model openai/gpt-oss-120b) with no fallback, rather than the application's shipped two-provider configuration. The results therefore characterise the model under this one serving configuration, not the full deployed system. Served provider and resolved model names were logged for every call.

Defences against measurement artefacts were enforced per call: calls were issued strictly sequentially, and a response served from cache or labelled as a rule-based fallback would have invalidated a run. Default-fill screening is reported at the level the preserved evidence actually supports. The zero-score check (`atsScore === 0`) was applied to all 49 preserved raw records and to the 31 accepted observations and found no hits: 0 zero scores, recorded in the observations file as `invalid_outputs_ats_zero: 0`. The empty-summary check could not be independently completed: none of the 49 preserved raw JSONL records contains summary text — the records store `atsScore`, `breakdown`, latency, attempt and provenance fields, but no summary field — so "empty summary" is not a condition that can be tested against the surviving evidence. Whether any accepted output carried an empty summary therefore remains unverified, and this is carried as a limitation (Section 6). No claim is made that every possible default-fill condition was ruled out.

3.2 Synthetic benchmark

All inputs are synthetic. Benchmark `careerlens-benchmark-v1` (version 1.0.0; SHA-256 `761f481acfbddffd28957e3531836e50c32762687ae10ce9641ba432312efc1c`) was generated offline by a seeded deterministic script (`research/scripts/benchmark_corpus_v1.mjs`, seed 20261013) and frozen before any analysed call was made. It comprises 12 synthetic resumes (350–600 words, a degree, a single skills line, at least two dated experiences) fully crossed with 10 synthetic job descriptions (each with a ten-term vocabulary and at least three must-have requirements), giving 120 resume–job pairs. Each job description is balanced at four high-, four mid-, and four low-overlap pairs by construction. The same benchmark reserves 12 strata with 48 controlled-edit variants (role-relevant additions and removals, an irrelevant-skill addition, and length-matched irrelevant prose) for the unexecuted RQ3 study. Table 1 summarises the frozen structure.

Table 1. Frozen benchmark `careerlens-benchmark-v1` (verified 2026-10-13; full per-pair validation in the project's data-validation records)

Property	Value
Resumes × job descriptions	12 × 10, fully crossed = 120 pairs
Resume properties	350–600 words; degree; single skills line; ≥2 dated experiences
Job-description properties	10-term vocabulary; ≥3 must-have lines; 5 education-flagged
Tier balance	4 high / 4 mid / 4 low per job description (all 10)
RQ3 strata (reserved, unused here)	12 cells; 48/48 controlled edits constructed and offline-verified
Integrity	100% synthetic texts; seeded generator; frozen checksum above

3.3 Data collection, run order, and ledger

Every provider attempt was appended to a call ledger (`research/logs/CALL_LEDGER.csv`) recording case, run index, purpose, outcome code, latency, served provider, and token counts; raw response records are preserved verbatim (`research/results/raw/groq_tierb_chains.jsonl`). Three distinct things must be kept separate: how the benchmark was ordered, how the analysed observations were selected, and what the panel is representative of.

Ordering within the benchmark. Pairs were issued strictly sequentially in a frozen randomised order. `research/logs/RUN_ORDER.json` records a seeded pseudorandom permutation (mulberry32 shuffle with Fisher–Yates swaps over the sorted pair identifiers, seed 20261013) of all 120 benchmark pairs, frozen before any analysed call was made. Failed attempts were never silently discarded: they remained in the ledger and were retried until a valid response was obtained for that pair.

Selection of the analysed observations. The 31 analysed observations are exactly the first 31 positions of that frozen ordering — verified position by position against `RUN_ORDER.json`: the analysed set equals positions 1–31 of the permutation, and no pair outside those positions was ever attempted. Within each pair, the analysed record is the

first valid response; retries after failures and later duplicate successes never changed which pairs are in the panel (11 failed attempts and 4 superseded duplicate successes remain in the audit trail). Collection stopped after these 31 pairs because further calls were not authorised, not because of anything observed in the data. The protocol's focused-study amendment — adopted after collection began but before any analysis was run — had additionally specified 12 repeat runs for RQ2 and 3 baseline plus 16 controlled-edit runs for RQ3; none of those calls was ever made (Section 4.3). Accordingly, this paper reports RQ1 alone.

Representativeness. Within the synthetic benchmark, panel membership was fixed by a random ordering frozen in advance and stopped for authorisation reasons, with no inclusion or exclusion based on scores. That is the whole of the guarantee: the panel inherits the benchmark's construction — 31 clustered pairs of synthetic resume and job texts — and a seeded ordering of synthetic pairs cannot make it representative of real-world resumes or job postings. No population-level or representativeness claim is made (Section 6).

Rule-arm scores were recomputed offline from the frozen texts, so no rule-arm call is involved. For the AI arm, the analysis uses the first valid success per pair; later duplicate successes are marked as superseded and excluded by a pre-specified rule, with all raw rows preserved. The analysis library was frozen and passed 191/191 code-level tests on synthetic fixtures (fixtures are code tests, not research data).

3.4 Statistical analysis

For each pair, the difference d is the AI score minus the rule score, on the 0–100 scale. Following the agreement-measurement tradition of Bland and Altman, the primary summary is the **mean absolute score difference**: the average absolute difference between the AI score and the rule-based score across the 31 observations — the mean of $|d|$ in score points. For traceability, the saved analysis files label this same statistic `MAE` (mean absolute error); that label is retained only in the archived files and in this sentence, because no ground-truth score exists against which an error could be defined. The **median absolute difference** (the 50th percentile of $|d|$, 8 points here) is a different quantity: it is reported alongside the mean as a robust counterpart, not as another name for it. We also report the mean bias (the average of d, indicating systematic offset), the SD of the differences, and the limits of agreement, defined as the mean bias ± 1.96 SD — the range within which approximately 95% of pairwise differences would fall if differences were approximately normal. Because operational decisions on a 0–100 scale are usually band-based, we additionally report the proportions of pairs within ±5 and ±10 points.

Correlation is reported only as association: Pearson r and Spearman ρ are computed, but correlation is explicitly not agreement — two scorers can correlate perfectly and still differ by a constant offset. Three cautions apply to all interval statements. First, the 31 pairs form a crossed design (12 resumes × 10 job descriptions), so pairs are not independent; the Fisher-z reference half-width ($1.96/\sqrt{(n-3)} = 0.37$ at $n = 31$) assumes independence and is reported as a reference only. Second, a two-way cluster bootstrap over resume and job clusters (10,000 replicates, seed 20261013) provides indicative ranges for the mean absolute score difference, median absolute difference, bias, the ±5/±10 proportions, and Spearman ρ; with at most 12 and 10 clusters these bootstrap intervals under-cover and are labelled indicative, never calibrated 95% coverage. Third, no ground truth exists, so no accuracy interval of any kind is computed.

3.5 Reproducibility

The frozen benchmark, run order, call ledger, raw response records, observations file, analysis library, and results file are retained in the project's local research archive with SHA-256 checksums (a 27-file manifest) and step-by-step rerun instructions. Three different claims must be kept apart.

Reproduction of the RQ1 analysis (successful). A clean-copy run — executed offline from a copy of the archived inputs, with network imports blocked, not from the working tree — regenerated all five RQ1 analysis outputs (the observations file, the results file, the human-readable summary, and both figure sources) byte for byte, and re-ran the 191/191 code-level tests with the same counts. A pre- and post-run hash snapshot confirmed that 15 of 15 original

files were unaltered, and a full manifest re-verification matched 27 of 27 checksums. Re-checked during preparation of this manuscript, the manifest still matches every frozen research input and output; only three living working documents — the journal-requirements log and the two literature records — had been edited after the manifest snapshot and no longer match their recorded hashes.

Independent verification (successful). Separately from reproduction, an independent verification script recomputed the reported statistics from the frozen observations using different numerical algorithms (Kahan summation, Welford, single-pass Pearson, rank-based Spearman, Fisher-z), agreeing to a tolerance of 1e-9 on all 15 checks. During preparation of this manuscript, every number reported here was additionally re-derived directly from the 31 saved observations and matched the saved results file.

Source-version pin (missing). The rule-based scores are recomputed offline by executing `buildMatchAnalysis` in `backend/ruleBasedFallback.js` on the frozen texts. The project has no source-control history — no Git repository exists — so no historical commit pins the backend source version, and the reproducibility manifest does not checksum the backend application file. The exact source revision that produced the stored rule scores therefore cannot be verified after the fact. This is a provenance limitation of the archive; it does not affect the reproduced analysis outputs, but it prevents an exact historical source-code claim.

Finally, none of this is publicly accessible: the archive exists only on the authors' local systems, and no external repository has been created. The data-availability statement in the Statements and Declarations reflects that.

4. Results

4.1 Execution status

Table 2 states what was actually collected. The 31 accepted baselines are the first valid response for each of the first 31 pairs in the frozen order; four later duplicate successes were superseded by the pre-specified first-valid rule; eleven failed attempts (six provider-unreachable, three bad requests, two timeouts) were retained in the ledger, retried, and never used as data; three further ledger rows came from a harness defect (zero HTTP requests issued), were recorded as such, and were corrected in place with their original timestamps preserved. On default-fill screening: no accepted output had a zero ATS score (0 of 31; saved audit field `invalid_outputs_ats_zero: 0`), while empty-summary screening could not be completed from the preserved records, which contain no summary text (Section 3.1). The ledger records 47 counted HTTP requests and 111,627 tokens in total. No further calls were made after the panel completed.

Table 2. Execution state of the pre-specified study components, verified 2026-10-14 from the ledger, raw records, and the observations file

Component	Planned	Valid observations collected
RQ1 baseline scores (first valid per pair)	31	31
RQ2 repeat runs (6 pairs × 2 runs)	12	0
RQ3 missing baselines (3 pairs)	3	0
RQ3 controlled-edit runs (4 pairs × 4 edits)	16	0
Ledger integrity	—	50 rows (1 preflight + 31 RQ1 + 18 retries); raw records 49; 11 failures retained; 4 duplicates superseded; 0 zero-ATS-score outputs; empty-summary check not possible from preserved records

4.2 RQ1: score agreement on the 31-pair panel

The panel spans all 12 resumes and 10 job descriptions and all three designed tiers (12 high, 10 mid, 9 low). Rule scores range from 20 to 84 and AI scores from 13 to 92 on the shared 0–100 scale.

Table 3. RQ1 agreement statistics for the 31-pair panel (computed from the archived observations file; indicative intervals from the two-way cluster bootstrap, 10,000 replicates; agreement is not accuracy)

Statistic	Value
Pairs, n	31
Mean absolute score difference (points)	9.13 (indicative interval 5.93–13.22)
Median absolute difference (points)	8 (indicative interval 4–12)
Mean bias, AI − rule (points)	+4.16 (indicative interval −1.32 to +10.18; crosses zero)
SD of differences (points)	10.49
Limits of agreement (points)	−16.40 to +24.72
Pairs within ±5 points	38.71% (12/31; indicative interval 13.3–66.7%)
Pairs within ±10 points	67.74% (21/31; indicative interval 39.3–92.0%)
Pearson r	0.91
Spearman ρ	0.91 (indicative interval 0.69–0.98)
Fisher-z reference half-width for r	0.37 (reference only; independence does not hold; interval 0.82–0.96)
Pairs excluded	0

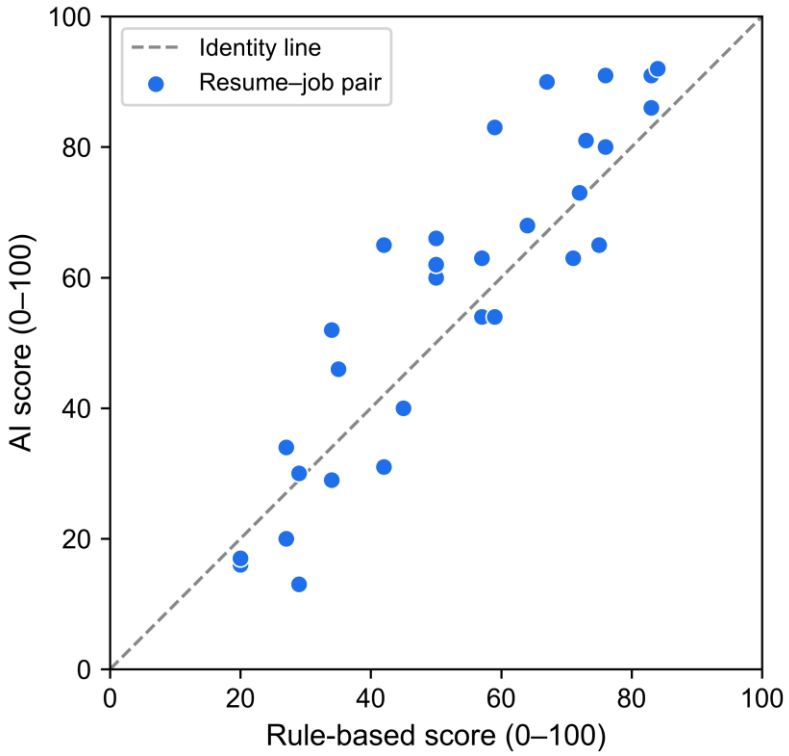


Fig. 1 *Scatter of AI scores against rule-based scores for the 31 resume–job pairs; the dashed line is the identity line of perfect agreement. Each point is the first valid baseline score of one pair (Table 4)*

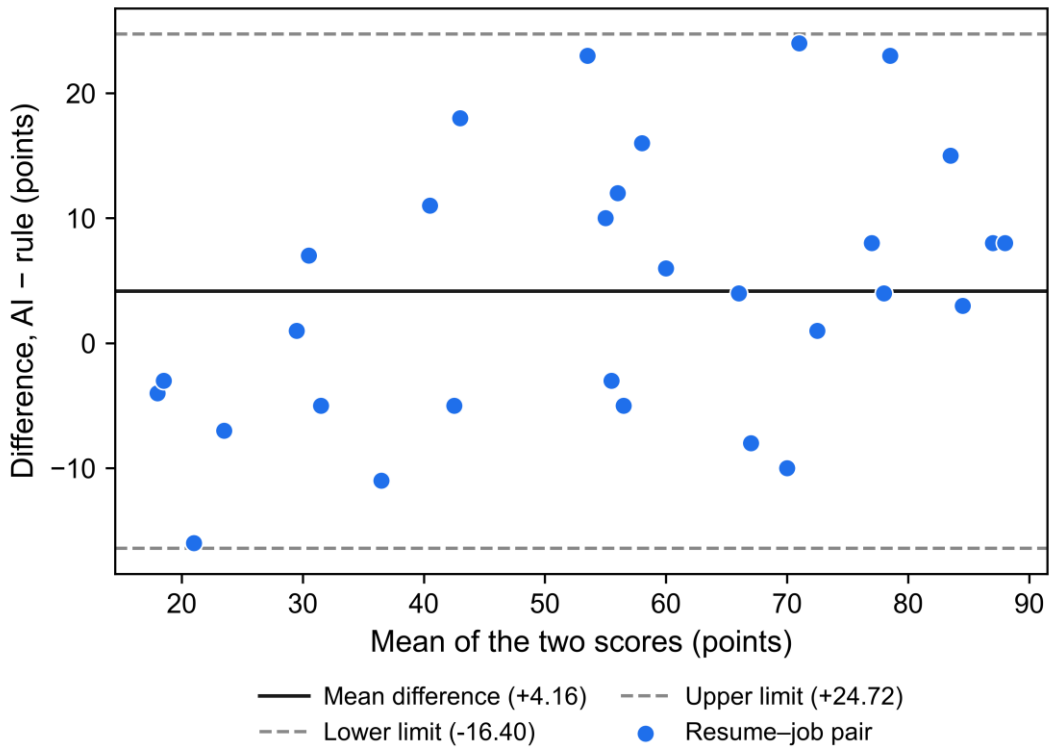


Fig. 2 *Bland–Altman plot of AI − rule score differences against the mean of the two scores for the same 31 pairs; the solid line is the mean difference (+4.16 points) and the dashed lines are the limits of agreement (−16.40 and +24.72 points)*

Fig. 1 shows that the two scorers order the panel similarly, with visible scatter around the identity line; Fig. 2 shows that differences are spread across the whole score range rather than concentrated at one end, with no obvious trend of the difference with the pair mean at this sample size. Table 4 lists every pair.

Table 4. Per-pair observations (rule scores recomputed offline from the frozen texts; AI scores are first-valid baselines; sorted by pair identifier)

Pair	Designed tier	Rule score	AI score
J1__R1	High	84	92
J1__R2	High	76	80
J1__R5	Mid	59	54
J1__R10	High	75	65
J2__R4	Mid	42	31
J2__R7	Low	34	29
J2__R8	Low	34	52
J2__R10	Mid	50	60
J2__R12	Mid	57	63
J3__R3	Low	29	13
J3__R4	High	83	91
J4__R1	High	71	63
J4__R3	Low	27	20
J4__R12	Mid	57	54
J5__R7	High	76	91
J5__R9	High	76	80

Pair	Designed tier	Rule score	AI score
J5__R12	Mid	45	40
J6__R1	High	72	73
J6__R8	High	73	81
J6__R11	Mid	50	62
J7__R12	High	83	86
J8__R2	High	64	68
J8__R3	Mid	42	65
J8__R8	Low	27	34
J8__R11	Mid	59	83
J9__R5	High	67	90
J9__R8	Low	29	30
J10__R3	Low	35	46
J10__R5	Low	20	17
J10__R6	Low	20	16
J10__R9	Mid	50	66

Three features of the panel deserve plain statement. First, the two scorers are strongly rank-associated (Pearson r = 0.91, Spearman ρ = 0.91), consistent with the pre-specified directional expectation; this is association under a crossed, non-independent design, not agreement and not precision. Second, absolute disagreement is material relative to the scale: the median gap is 8 points, the mean absolute score difference is 9.13 points, and the limits of agreement span roughly 41 points. The largest observed difference is **+24 points with the AI score higher** (pair J8__R11: AI 83, rule 59); the largest difference with the AI score lower is **−16 points** (pair J3__R3: AI 13, rule 29), so observed gaps do not reach 20 points in that direction. Six of the 31 pairs (19.35%) have absolute differences of 16 points or more. Third, the AI scores are on average 4.16 points higher, but the indicative interval for the bias crosses zero, so these data do not establish a systematic offset; they establish only the in-panel average.

4.3 Pre-specified questions without observations

RQ2 and RQ3 were designed but never executed (Table 2): 0 of 12 planned RQ2 repeat runs and 0 of 3 planned RQ3 baselines plus 0 of 16 planned RQ3 edit runs exist. No repeatability statistic, sensitivity delta, direction count, or significance test exists for either question, and none is presented; the zeros in Table 2 are counts of missing observations, not measurements of an absent effect. The rule arm's pre-specified RQ3 expectations (strictly positive or negative deltas for role-relevant edits, exactly zero for irrelevant edits) were verified against the scoring code offline; they remain code-derived predictions of the deterministic engine, not empirical findings. Any statement in this paper about stability or edit sensitivity would be unsupported, so none is made.

5. Discussion

What the panel shows. On 31 synthetic resume–job pairs, a keyword-based rule engine and an LLM scorer under one fixed configuration agree closely on the ordering of pairs (ρ = 0.91) but disagree on the numbers: a mean absolute score difference of 9.13 points (median 8), with six of the 31 pairs (19.35%) differing by 16 points or more and only 38.71% within ±5 points. This combination — high correlation, material pairwise disagreement — is the classic reason agreement methods exist: correlation alone would have suggested interchangeability that the differences do not support. The pattern also echoes the wider literature: LLM screening scores have been reported to diverge sharply from human ratings [4] and to swing with superficial formatting [10]; here, even against a transparent rule baseline on identical inputs, single-number summaries understate pair-level spread.

Relation to prior work. Compared with the LLM-versus-human studies [4, 6, 7], this study moves the comparison to two automated systems on identical inputs — a setting in which neither side can appeal to human judgement, and in which a deterministic control makes exact-zero expectations provable from code. Compared with the constructed-ground-truth validity audit [15], the construct here is deliberately weaker and honestly labelled: correspondence

between two scorers, not accuracy against labels. Compared with the cross-system extraction comparison [13] and the cross-context repeatability preprint [14], the contribution reported here is the shared-scale score comparison; the repeatability and controlled-edit components of our design remain unmeasured and are offered as a pre-specified protocol rather than as results.

Operational reading. For teams running a dual-scorer pipeline of this kind, the panel supports one modest practice suggestion: log both scores and their difference, and treat a ±5 or ±10-point band policy as an explicit operational choice, because the two numbers are not substitutes for one another on individual pairs. This is a feasibility observation from 31 synthetic pairs under one configuration, not a deployment recommendation, and no claim is made that either scorer is closer to any notion of correctness.

Scope and significance. The evidence base is deliberately narrow: 31 synthetic pairs, one provider, one model, one prompt, no ground truth, and no repeatability or sensitivity data. Whether a transparent, reproducible agreement baseline plus a pre-specified protocol for the harder questions is sufficient for a full research article is a judgement for reviewers and for the authors' faculty review; the paper is written so that this judgement can be made on stated evidence rather than on presentation.

6. Limitations and threats to validity

1. **Two of three questions are unmeasured.** The protocol's other questions — run-to-run repeatability and sensitivity to controlled edits — have no data at all (RQ2: 0/12; RQ3: 0/3 and 0/16). This is an RQ1 report, not the three-question study the protocol describes; the post-initiation narrowing of reporting scope is disclosed in Sections 1.2, 3.3, and 4.3, and the protocol itself is unchanged. The missing observations are not findings: they provide no evidence either way about repeatability or edit sensitivity.

2. **Small, clustered, synthetic panel.** The 31 pairs are the first 31 positions of a seeded random ordering of the benchmark, frozen before collection; membership was never determined by the observed scores, but neither that ordering nor the benchmark's synthetic construction yields a panel that is representative of real resumes or job postings, and no such claim is made. The crossed resume-by-job design violates the independence assumptions behind conventional intervals, which is why the Fisher-z interval is reference-only and the cluster-bootstrap intervals are labelled indicative. Synthetic, construction-passing texts carry no real-world resume distribution, no demographic variation [5, 9], and no format variation [10].

3. **Single provider, model, and prompt.** The AI arm was served by one provider with no fallback (a recorded deviation from the shipped two-provider configuration); the model is addressed through a rolling server-side alias, and sampling at temperature 0.1 is stochastic. The single fixed prompt bounds generalisability, since prompt formatting alone can move LLM behaviour substantially [10].

4. **No ground truth.** Every statistic here measures correspondence between two scorers; weak or strong values would be equally non-validating of either scorer's accuracy [11].

5. **Default-fill screening is incomplete.** The zero-score check could be run against the preserved records and found nothing (0 of 31 accepted observations); the empty-summary check could not be run at all, because summary text is absent from the preserved raw JSONL. The surviving evidence therefore cannot exclude every default-fill condition.

6. **Rule-score provenance is not version-pinned.** Rule scores were recomputed offline from the current backend source; with no source-control history and no checksum of the backend file in the manifest, the exact historical source revision cannot be established (Section 3.5).

7. **Collection stopped early.** Further model calls were not authorised once verified provider-quota evidence was unavailable; the outstanding pre-specified runs remain unexecuted, and the reported panel is bounded by that constraint.

8. **Targeted, not systematic, literature review.** The related-work search used no database strategy, PRISMA flow, or dual screening; the gap statement is qualified accordingly and must be re-verified against a broader search before submission.

7. Conclusion and future work

On a frozen 31-pair synthetic panel, the AI and rule-based CareerLens scorers were strongly rank-associated (Pearson $r = 0.91$, Spearman $\rho = 0.91$) yet showed a mean absolute score difference of 9.13 points and a median absolute difference of 8 points, with a mean bias of +4.16 points and limits of agreement from −16.40 to +24.72 points; 38.71% of pairs lay within ±5 points and 67.74% within ±10. The defensible reading is narrow: within this panel and configuration, the two scorers agree on ordering but not on individual numbers, and no accuracy, fairness, or generalisability claim follows — agreement is not validity.

The natural next steps are the pre-specified but unexecuted studies: the 12 planned repeat runs that would answer the repeatability question, and the 19 planned baseline and controlled-edit runs that would answer the sensitivity question, each now fully specified in the archived protocol. Beyond that, replication with multiple models and prompts, and with real rather than synthetic resume distributions, would test whether the pattern observed here — strong association with material pairwise spread — holds more broadly, as would a systematic literature search to firm up the gap statement.

Statements and Declarations

Competing interests. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. [Proposed wording; each author must confirm or amend before submission.]

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Ethics approval and consent. This study uses entirely synthetic resumes and job descriptions generated by a documented, seeded script; it involves no human participants, no personal data, and no collection of real applicant records. [No institutional ethics determination, exemption, or waiver has been obtained, and none is claimed. Before submission the authors must confirm with their institution whether a formal determination or exemption statement is required for this work, and the confirmed wording must replace this placeholder; until that confirmation exists, this statement remains provisional.]

Data and code availability. All data supporting the RQ1 findings are within the paper — including Table 4, which lists every observation — its figures, and the locally archived analysis files: the frozen benchmark `research/data/benchmark-v1/benchmark_v1.json` (SHA-256 `761f481a…`), the call ledger `research/logs/CALL_LEDGER.csv`, the raw attempt records `research/results/raw/groq_tierb_chains.jsonl`, the RQ1 observations `research/results/analysis/rq1_observations_v1.json`, the computed metrics `research/results/analysis/rq1_results_v1.json`, the analysis library `research/scripts/analysis_core.mjs`, and the protocol and literature records at the project root; a checksummed reproducibility package (27-file manifest, reproduction run log, secret scan, rerun instructions) accompanies the archive locally. These files are held only on the authors' local systems and are not published anywhere. [REPOSITORY LINK — BLOCKED: no external repository exists; no URL is given because none has been created. The authors must first approve an archival location, create the repository, and confirm the access route (proposed corresponding author, to be confirmed) before this statement can name one. Rule-based scores were recomputed from the current backend source without a version pin (Section 3.5).]

Authors' contributions. [A CRediT-style contribution statement is to be completed and confirmed by all authors before submission; no contributions are asserted on the authors' behalf in this draft.]

Use of generative AI in manuscript preparation. Generative AI tools were used in preparing this manuscript, and the use is disclosed here in line with the publisher's AI policy. An AI research and writing assistant (OpenCode agent framework; the exact underlying model designation is to be confirmed by the authors at submission) assisted with substantive work: structuring the manuscript, drafting and restructuring text, implementing analysis and verification scripts, and rendering figures from the frozen observation file — assistance that the publisher's framework classifies as requiring explicit declaration and author accountability, not mere copy-editing. The AI systems studied in the research itself are documented in Section 3.1. All quantitative values in this paper were read from frozen, human-inspectable files (benchmark, ledger, raw responses, observations) and were independently recomputed from the saved observations during preparation of this revision; no AI-generated data are presented as results, no AI-generated figures depict anything not present in the archived data, and no AI tool is an author. The authors review and take full responsibility for all content. [This disclosure must be verified and approved by all authors before submission.]

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